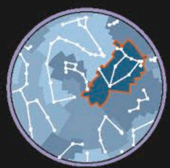
CEPHEUS

Cepheus



SIZE RANKING 27

BRIGHTEST STAR Alpha (α) 2.5

GENITIVE Cephei

ABBREVIATION Cep

HIGHEST IN SKY AT 10 PM September–October

FULLY VISIBLE 90°N–1°S

Cepheus lies in the far northern sky between Cassiopeia and Draco. Its main stars form a distorted tower or steeple shape, yet this ancient Greek constellation in fact represents the mythical King Cepheus of Ethiopia, who was the husband of Queen Cassiopeia and the father of Andromeda. Cepheus is not a particularly prominent constellation.

SPECIFIC FEATURES

The constellation's most celebrated star is Delta (δ) Cephei (see p.286), from which all Cepheid variables take their name. Just under 1,000 light-years away, this yellow-colored supergiant varies between magnitudes 3.5 and 4.4 every 5 days 9 hours.

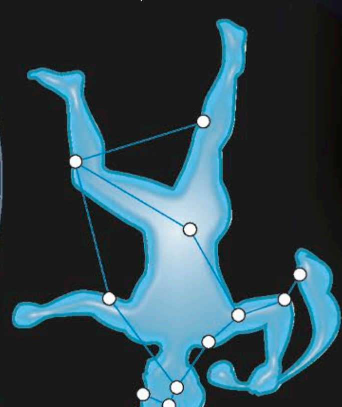
IC 1396
The Garnet Star or Mu Cephei (top left) lies on the edge of the large but faint nebula IC 1396. The nebula is centered on the 6th-magnitude multiple star Struve 2816.

These changes can be followed with the naked eye. Delta (δ) Cephei is also a double star; its 6th-magnitude, blue-white companion is visible through a small telescope.

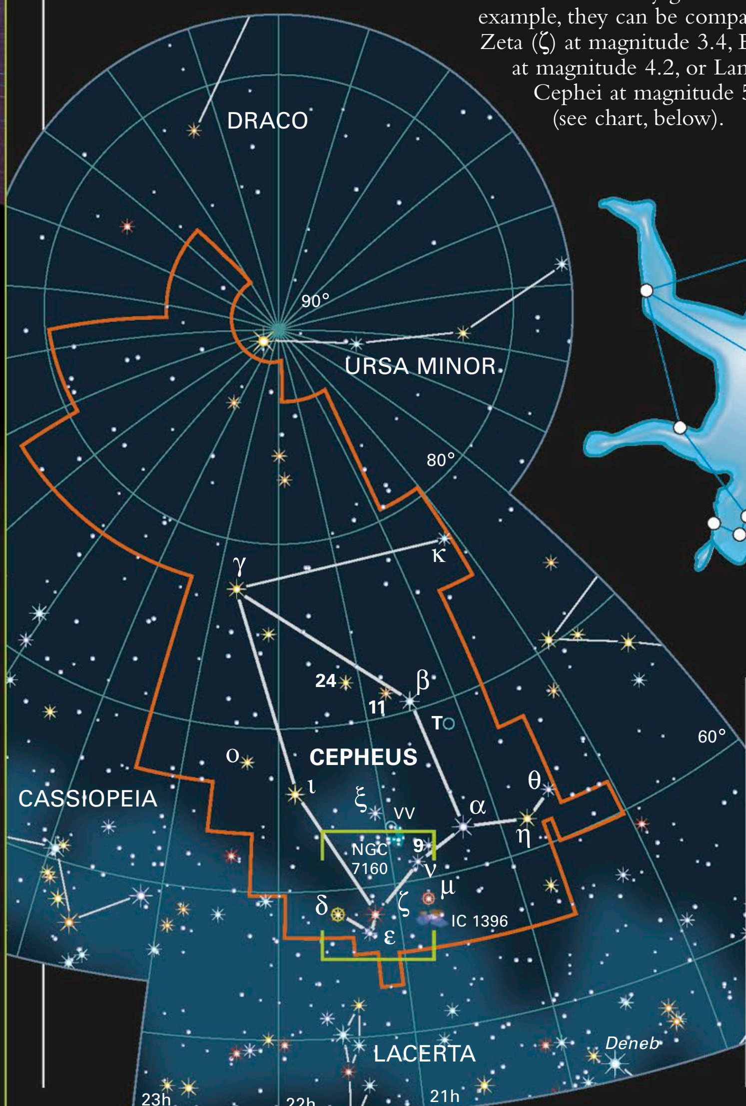
A significant variable star of a different kind is Mu (μ) Cephei, which is a red supergiant that ranges anywhere between magnitudes 3.4 and 5.1 every 2 years or so. This supergiant is also known as the Garnet Star on account of its strong red coloration.

Nonvariable stars near Delta (δ) and Mu (μ) Cephei can be used to gauge the magnitude of these two variable stars at any given time. For example, they can be compared to Zeta (ζ) at magnitude 3.4, Epsilon (ε) at magnitude 4.2, or Lambda (λ) Cephei at magnitude 5.1 (see chart, below).

THE KING
Shaped like a bishop's miter, Cepheus is not easy to pick out in the sky. He is flanked by his prominent wife, Cassiopeia, and Draco, the dragon.



CEPHEUS



DELTA (Δ) AND MU (Μ) CEPHEI

MAGNITUDE KEY

0.0–0.9
1.0–1.9
2.0–2.9
3.0–3.9
4.0–4.9
5.0–5.9
6.0–6.9

HENRIETTA LEAVITT

Henrietta Swan Leavitt (1868–1921) worked at Harvard College Observatory in the early 20th century. Her study of variable stars in the Small Magellanic Cloud led to the period-luminosity law. This law links the variation period of a Cepheid variable to its intrinsic brightness, which in turn can indicate distance. Her law remains fundamental to our knowledge of the scale of the universe.

DETERMINATION
By painstakingly measuring photographic plates, Henrietta Leavitt discovered 2,400 variable stars of all types.

THE NIGHT SKY